KARTIKE CHAURASIA

(716) 907-3496 | kartike2001@gmail.com | Website: Kartike | LinkedIn: Kartike | GitHub: Kartike

WORK EXPERIENCE

Site Reliability Engineer – Identity and Access Management, *Coinbase* — ***September 2024 – Present***

- Modernized a scalable authorization platform replacing a legacy LDAP-based access control system, supporting 200M+ authorization checks/month and reducing access latency by up to 88x while maintaining operational continuity
- Upgraded a critical identity service from Tier-3 to Tier-1 by meeting 99.99% uptime and P99 latency <150ms targets through smoke tests, failover drills, and production-readiness validation by increasing test coverage from 5% to 90%
- Automated self-healing and health monitoring workflows, preventing 7+ high-severity incidents and saving 40+ on-call engineering hours annually
- Diagnosed failing production health checks by tracing logs, service probes, and configuration behavior, restoring correct restart behavior and improving service resilience
- Designed and implemented a fine-grained authorization framework enabling field-level access control via JWT-based authentication, enforcing least-privilege policies across APIs and establishing a new standard for granular authorization within Coinbase services
- Established a cross-IdP authentication bridge that enabled SSO applications to securely access backend services with full user-level permission enforcement, eliminating proxy-based workarounds that bypassed the authorization layer
- Engineered a new Golang ingestion and client framework, reducing onboarding time for new integrations from 2 months to 1 day, eliminating 2,000+ lines of redundant code, and improving maintainability
- Implemented and released a Kafka-backed real-time event stream for identity changes, enabling downstream services to consume field-level updates and migrate off a legacy event pipeline
- Developed an AST-based SCIM filter parsing and query pipeline supporting complex nested queries and 50+ field mappings, enabling advanced identity search capabilities and vendor-agnostic integrations
- Implemented wallet-based, cryptographically secure identity verification for self-service account and device unlocks, eliminating 650+ annual IT support tickets and reducing mean resolution time from hours to minutes
- Architected secure identity lifecycle enforcement for wallet-based identity verification, automatically revoking and delinking wallets upon identity deactivation to prevent unauthorized access and reduce residual credential risk
- Developed a Model Context Protocol server for a unified data service aggregating identity and HR data from SaaS providers such as Okta and Workday, leveraging the SCIM-based API to simplify cross-system data queries
- Migrated the database from Neptune to PostgreSQL, simplifying local development, improving reliability, reducing operational cost, and enabling transactional consistency for identity workflows

Head Teaching Assistant – Data Structures & Algorithms, *University at Buffalo* — ***August 2022 – May 2024***

- Developed an automated testing system using Python scripting and Java, streamlining the coding assignments evaluation process for over 400 students and employing JSON for test result storage along with Makefile for build automation

Full Stack Web Developer, *University at Buffalo* — ***August 2022 – May 2024***

- Developed a Python-based TCP Handler (backend) for managing incoming requests and TCP Packets with MongoDB integration, built ADA-compliant web interfaces using HTML, CSS, and JavaScript, employing Agile methodologies
- Leveraged WebSocket to create a live, real-time scoreboard, reducing workload by up to 90% compared to the prior conference, initially joined as a volunteer but transitioned to a part-time role based on performance and contributions

Software Engineering Intern, *Qarik Group* — ***June 2022 – August 2022***

- Worked within a team of 4 to streamline the deployment, scaling, and upgrading of systems like Cloud Foundry across multiple CI/CD pipelines enhancing the security and efficiency of web application components
- Developed Go (Gin) APIs to store and update deployment records in a PostgreSQL container, added React features and integrated deployment data via the GitHub API

SKILLS

Programming & Scripting: Go (Golang), Python, Java, Scala, C, Bash/Shell Scripting
Infrastructure & Cloud: Kubernetes, Docker, Linux/Unix Systems, TCP/IP Networking, Load Balancing, DNS, TLS, Reverse Proxies
Automation & CI/CD: Git, GitHub Actions, Infrastructure as Code (Terraform), Helm
Databases & Storage: PostgreSQL, MongoDB, Redis, Neptune, Neo4j
Testing & Quality: Unit Testing, Integration Testing, Load/Performance Testing, Automated Smoke Tests

EDUCATION

University at Buffalo, The State University of New York — ***May 2024***
B.S. Computer Science; Minor: Mathematics — **GPA: 3.84/4.00**

- Tau Beta Pi - Engineering Honor Society
- Phi Beta Kappa - Academic Honor Society
- 2022 Leaders in Excellence Award
- 2023 Innovative Student Leader Award